

Errors & Data Formats

Post-Assessment 3 · 8 questions · 30 points

Name: _____

Date: _____

Score: / 30

This is a **post-assessment**. It shows what you learned in this unit. Try every question, and check your answers if you have time.

Work on your own, without a computer. You have about **40 minutes**. When a question asks what a program prints, write exactly what Python would print, one line of output per answer line. If the code would crash, write the name of the error (for example, `TypeError`). Show your thinking; partial credit is given.

1. Which exception?

4 points

Each line crashes. Write the exception name: **ValueError**, **ZeroDivisionError**, **KeyError**, or **FileNotFoundError**. Each one is used once.

Code	Exception
<code>open("no_such_file.csv")</code>	
<code>{"Ana": 9}["ana"]</code>	
<code>10 // 0</code>	
<code>float("3.5.1")</code>	

2. Read the traceback

3 points

TRACEBACK

```
Traceback (most recent call last):
  File "pet_registry.py", line 12, in <module>
    age = get_age(pets, "Rex")
  File "pet_registry.py", line 5, in get_age
    return pets[name]["age"]
KeyError: 'Rex'
```

(a) What type of error happened?

(1 pt)

(b) On which line number did the error actually happen?

(1 pt)

(c) On which line number was the function called?

(1 pt)

3. Safe Divider

4 points

Write exactly what this program prints.

```
for text in ["abc", "5", "0"]:
    try:
        n = int(text)
        print(100 // n)
    except ZeroDivisionError:
        print("No zeros!")
    except ValueError:
        print("Digits only")
    else:
        print("OK")
    finally:
        print("----")
```

4. Wrong net

2 points

```
scores = {"Ana": 90}
try:
    print(scores["Ben"])
except ValueError:
    print("No score for that name")
```

(a) What happens when this runs?

(1 pt)

(b) Fix the program so it prints `No score for that name` instead.

(1 pt)

5. Bulletproof input

4 points

```
def ask_age(prompt):  
    while True:  
        try:  
            age = int(input(prompt))  
        except ValueError:  
            print("Numbers only.")  
            continue  
        if _____:  
            return age  
        print("Age must be 1 to 120.")
```

(a) Fill in the blank so only ages from 1 to 120 (including 1 and 120) are accepted.

(2 pts)

(b) The user types `ten`, then `200`, then `12`. What does the function print, and what does it return?

(2 pts)

6. Quiz Question Bank

4 points

FILE: QUIZ.CSV

```
question,answer,points
2+3,5,1
Capital of Japan,Tokyo,2
Opposite of hot,cold,1
Largest ocean,Pacific,3
```

```
import csv
with open("quiz.csv") as f:
    rows = list(csv.reader(f))
print(len(rows))
print(rows[2])

with open("quiz.csv") as f:
    for row in csv.DictReader(f):
        if int(row["points"]) >= 2:
            print(row["answer"])
```

(a) Write exactly what the program prints.

(3 pts)

(b) Why does the program need `int(row["points"])`?

(1 pt)

7. Quiz Question Bank

4 points

```
import json
bank = [{"q": "5 x 6?", "a": "30"}, {"q": "Color of the sky?", "a": "blue"}]
with open("bank.json", "w") as f:
    json.dump(bank, f)

with open("bank.json") as f:
    loaded = json.load(f)
print(len(loaded))
print(loaded[1]["a"])
```

(a) What does the first `print` show? (1 pt)

(b) What does the second `print` show? (1 pt)

(c) What Python type is `loaded`? (1 pt)

(d) A Python **dictionary** is saved in JSON as a(n) _____. (1 pt)

8. Load the high score

5 points

Write a function `load_high_score(filename)` that reads a file containing one whole number and **returns** it as an `int`. If the file does not exist, it should print `No high score yet` and return `0` instead of crashing.

END OF ASSESSMENT